

Write a C program to represent a graph using an adjacency matrix and perform Breadth First Search (BFS) traversal starting from a user-specified vertex. Display the order in which the vertices are visited during the traversal.

```

1 #include <stdio.h>
2
3 int main() {
4     int n, start;
5     scanf("%d", &n);
6
7     int a[n][n];
8     for (int i = 0; i < n; i++)
9         for (int j = 0; j < n; j++)
10             scanf("%d", &a[i][j]);
11
12     scanf("%d", &start);
13
14     int vis[n];
15     for (int i = 0; i < n; i++) vis[i] = 0;
16
17     int q[n], f = 0, r = 0;
18     vis[start] = 1;
19     q[r++] = start;
20
21     while (f < r) {

```

Your INPUT goes here!

OUTPUT

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Breadth First Search

BREADTH FIRST SEARCH TRAVERSAL

Write a C program to represent a graph using an adjacency matrix and perform Breadth First Search (BFS) traversal starting from a user-specified vertex. Display the order in which the vertices are visited during the traversal.

15for (int i = 0; i < n; i++) vis[i] = 0;

16

17int q[n], f = 0, r = 0;

18vis[start] = 1;

19q[r++] = start;

20

21while (f < r) {

22int u = q[f++];

23printf("%d ", u);

24

25for (int v = 0; v < n; v++) {

26if (!vis[v] && a[u][v]) {

27vis[v] = 1;

28q[r++] = v;

29}

30}

31}

32

33return 0;

34}

OUTPUT

Your input goes here!